Ram-Equivalent Local Certificates for the Kerr–Newman Penrose Trace System

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M513

October 8, 2026
Confidential: OpenAI722 comparative study, Batch25

Abstract

We examine the actual scalar trace equation in the pinned axisymmetric electrovacuum manuscript M513, separately from its divergence-form companion. We derive smooth coefficient variations, including the implicit graph parameter, and prove a residual comparison bound with unit constant for the rescaled height. We supply a drift-correct weak formulation, a conditional continuum inverse bound, a boundary-preserving Ram-equivalent root certificate, and charge–angular-momentum error budgets on the physical area branch. All local proofs are given with their domains and quantitative hypotheses. The electromagnetic correction to angular momentum, the source's order of limits, and the strict versus extremal equality distinction are retained. No candidate for the full geometric exterior is certified, and no independent global Penrose or original-data rigidity proof is claimed. Contribution assessment: C2/5, a little contribution through source-specific local residual, derivative, and branch certificates, without a demonstrated global gap closure, historical novelty, or measured computational improvement.

1 Scope, Source Assembly, and Attribution

The object of comparison is OpenAI's October 5, 2026 manuscript *The Kerr–Newman Penrose Inequality for Axisymmetric Electrovacuum Exteriors* [1], pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its L4 catalogue rating measures applicability, not realized contribution or validity. This report does not inherit an assessment from M510, M511, M514, or M517.

The source is not just `solve.tex`. Its `build/main.tex` assembles `history`, `axial`, `cut`, `comparison`, `reduction`, `deformation`, `ceiling`, `regularity`, `solve`, `adjoint`, and `rigidity` from `build/sections/`, and uses `build/references.bib`. We read the target solve and its coefficient definitions and relevant proofs, the conservation and model-parameter arguments, the reduction argument, and the local regularity mechanism. Other global components are dependency obligations, not independently certified theorems of this report. Assembly inventory is not proof validation.

The dated upstream audit selects original ID 513 and records all 15 public blobs unchanged between the pin and `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, with the same current edition directory. The accompanying M513 audit checks the actual local selected text files against the cached Git trees before relying on this statement. It is an observation at those commits, not a

fresh moving-head guarantee. Fourteen text files are locally available; the upstream PDF blob is represented in the tree, not treated as a locally read PDF.

Classical baseline. Maximum principles, Lax–Milgram, Dirichlet Green inverses, Cordes estimates, Miranda–Talenti, and the Banach contraction theorem are classical. Ram coordinates do not create these theorems. Our transfer uses actual local proofs in Astra TR8 (boundary-realized PDE/Green equivalence), Astra TR6 (complete state variation), and Astra TR1's `stability.tex` (residual-ball and coordinate-transfer proofs); see Section 14. The contribution is the checked specialization to the particular M513 coefficients and conservation variables.

Public literature checked on October 8, 2026. Khuri–Sokolowsky–Weinstein [2] work with maximal data and retain a horizon-rod correction; their Theorem 1.1 and Corollary 1.2 are not the unrestricted nonmaximal assertion of M513. The area–angular-momentum–charge theorem of Gabach Clement–Jaramillo–Reiris [3] has its own stability hypotheses. M513 instead explicitly assumes its physical area branch. Feng et al. [4], in their 2025 version, consider quasi-stationary vacuum data. Xu's current preprint [5], version 4 of January 2, 2026, claims an angular-momentum Penrose result using a modified Hawking functional; this report does not independently validate or import that claim. Smears–Süli [6] provide the classical nondivergence/Cordes framework. Public queries contained public mathematical terms only; no private manuscript or passage was transmitted. This is a bounded literature check, not a novelty or priority determination.

2 The Geometric Contract

In the source, Ω is a smooth oriented exterior diffeomorphic to the complement of a ball in $\mathbb{R}^3$, with connected compact boundary S and one asymptotically flat end. A smooth effective circle action of period 2π preserves $g, K, E_{\mathrm{f}}, B_{\mathrm{f}}, S$; η is its oriented generator. With $\tau = \mathrm{tr}_g K$, the sign conventions are

$$\begin{aligned} R_g + \tau^2 - |K|^2 &= 2(|E_{\mathrm{f}}|^2 + |B_{\mathrm{f}}|^2), \\ \operatorname{div}(K - \tau g) &= 2(E_{\mathrm{f}} \times B_{\mathrm{f}})^\flat, \qquad \operatorname{div} E_{\mathrm{f}} = \operatorname{div} B_{\mathrm{f}} = 0. \end{aligned} \tag{2.1}$$

The source assumes $g - \delta = O_4(r^{-1})$, $K = O_3(r^{-3})$, Coulomb leading fields with $O_3(r^{-3})$ remainders, the specified finite ADM/charge limits, and integrable energy and momentum densities. It sets $P_{\mathrm{ADM}} = 0$, $m = E_{\mathrm{ADM}} > 0$, and $Q^2 = Q_e^2 + Q_b^2$. No maximality assumption $\tau = 0$ is made.

The boundary is a future MOTS, $H_S + \mathrm{tr}_S K = 0$, with normal toward infinity, and satisfies the source's outermostness condition against every smooth enclosing weakly future trapped surface in the interior. In addition it is outer area-minimizing against every enclosing cut, including nonsymmetric and disconnected cuts, so its area A is the minimum enclosing area. The further explicit assumption is

$$A \geq 4\pi\sqrt{Q^4 + 4J^2}. \tag{2.2}$$

The source's claimed conclusion is

$$m^2 \geq \mathcal{P}(A, Q_e, Q_b, J) := \frac{A}{16\pi} + \frac{Q_e^2 + Q_b^2}{2} + \frac{\pi((Q_e^2 + Q_b^2)^2 + 4J^2)}{A}. \tag{2.3}$$

We study obligations entering this assertion, not a replacement proof of it.

On the strict branch $A > 4\pi\sqrt{Q^4 + 4J^2}$ the source claims equality characterizes the *original* data as an admissible spacelike exterior slice of subextremal dyonic Kerr–Newman, smoothly attached to

a future horizon section or the bifurcation sphere. Equality classification at the closed extremal endpoint is explicitly not asserted. A compact boundary must not be silently replaced by an extremal cylindrical end.

3 Conservation and Its Defect Budget

For $Q > 0$ use the constant duality rotation

$$E = \frac{Q_e E_{\mathrm{f}} + Q_b B_{\mathrm{f}}}{Q}, \qquad B = \frac{-Q_b E_{\mathrm{f}} + Q_e B_{\mathrm{f}}}{Q};$$

for $Q = 0$ leave the fields unchanged. The rotated charges are $(Q, 0)$; $|E|^2 + |B|^2$ and $E \times B$ are unchanged. The source topology and zero magnetic period give an invariant primitive α with $\mathrm{d}\alpha = *B^\flat$ and Cartesian decay $O_2(r^{-2})$. This primitive is not assumed for the original field when $Q_b \neq 0$. Write $\chi_{\mathrm{em}} = \alpha(\eta)$, distinct from the ellipticity coefficient used below, and set

$$\mathscr{J} = (K - \tau g)(\eta, \cdot)^\sharp + 2\chi_{\mathrm{em}} E, \qquad J(\Sigma) = \frac{1}{8\pi} \int_\Sigma \mathscr{J} \cdot \nu \, \mathrm{d}A_g. \tag{3.1}$$

Proposition 3.1 (Exact and defective conserved fluxes)**.** *Suppose η is Killing and α is invariant with the stated primitive relation. Define $d_E = \operatorname{div} E$, $d_B = \operatorname{div} B$, and $\mathfrak{r} = \operatorname{div}(K - \tau g) - 2(E \times B)^\flat$. For nested enclosing surfaces bounding a compact shell U, both normals toward infinity,*

$$Q_E(\Sigma_2) - Q_E(\Sigma_1) = \frac{1}{4\pi} \int_U d_E, \tag{3.2}$$

$$J(\Sigma_2) - J(\Sigma_1) = \frac{1}{8\pi} \int_U \big[\mathfrak{r}(\eta) + 2\chi_{\mathrm{em}} d_E\big]. \tag{3.3}$$

The charge formula also holds for B. In particular, exact electrovacuum data have conserved charges and conserved total *J. For approximate data the absolute integral of the displayed density bounds the corresponding flux error. The J assertion requires an exact primitive; a defect in that relation contributes an additional Cartan-identity error.*

Proof. Cartan's formula gives $\mathrm{d}\chi_{\mathrm{em}} = -\iota_\eta(*B^\flat)$ and $(E \times B) \cdot \eta = -E \cdot \nabla \chi_{\mathrm{em}}$ in the source's orientation. Contracting a symmetric tensor with $\nabla\eta$ gives zero, because η is Killing. Thus

$$\operatorname{div} \mathscr{J} = 2(E \times B) \cdot \eta + \mathfrak{r}(\eta) + 2E \cdot \nabla\chi_{\mathrm{em}} + 2\chi_{\mathrm{em}} d_E = \mathfrak{r}(\eta) + 2\chi_{\mathrm{em}} d_E.$$

The divergence theorem, with the inner boundary sign reversed, proves both formulas and their absolute-value bounds. At infinity $\chi_{\mathrm{em}} = O(r^{-1})$ and $E \cdot \nu = O(r^{-2})$, so the electromagnetic surface contribution is $O(r^{-1})$ and tends to zero. If the primitive changes by a closed invariant one-form β, Cartan gives $\mathrm{d}(\beta(\eta)) = 0$; its constant is zero on the axis. Hence the total flux is gauge-independent and equals J_{ADM}. □

This proof does not conserve the bare gravitational surface integral. The source potential normalization is

$$(\psi, \chi_{\mathrm{em}}, z) = (-Q, 0, -2J), \quad (Q, 0, 2J)$$

on the two oriented axis faces. The electromagnetic jump is $2Q$ and the twist jump is $4J$. Keeping these potentials fixed in the neutral metric deformation keeps these parameters in the comparison problem; it is not an assertion that each intermediate metric and tensor solves the original Einstein–Maxwell constraints.

4 The Actual Trace Operator

Fix prepared data, a smooth finite truncation Ω_R, $\varepsilon > 0$, $n > \max(4, 2/\varepsilon)$, and the source's smooth cutoff ϑ. Put

$$p(t) = n\vartheta(nt), \quad l(t) = \exp\Big(\int_0^t p(s)\,\mathrm{d}s\Big), \quad L(t) = l(t)e^{2t}, \quad \ell = e^{-n\varepsilon}, \quad \tau_0 = \ell^{3/2}. \tag{4.1}$$

Thus $0 \le p \le n$, $l' = pl$, $l = e^{nt}$ for $t \le 0$, and $l \le e$. For $P = \nabla f$ and $H = \operatorname{Hess} f$ introduce notation not conflicting with the area or model parameters:

$$D = 1 + l^2|P|^2, \quad d = D^{-1}, \quad b = \frac{l}{\sqrt{D}}, \quad w = bP, \quad v = |w|^2 = 1 - d, \tag{4.2}$$

$$Z = t + \tfrac{1}{8}\log D, \quad M = 4 + pv, \quad \chi = \frac{4d}{M}, \quad \beta = \frac{p+4}{M}, \quad \mathsf{A} = I - \beta w \otimes w, \quad u = L\sqrt{D}. \tag{4.3}$$

Here b denotes the trace multiplier only; the subextremality parameter later is b_*. Since $\partial_t(t + \frac{1}{8}\log D) = M/4 \ge 1$, the implicit equation defines a unique smooth $t = t(Z, P)$ for all finite Z, P. Its range is all $\mathbb{R}$ at fixed P. The eigenvalues of A are $1, 1, \chi$ and $0 < \chi \le 1$. The tensor formula, not $P/|P|$, defines it at zero slope.

At the physical endpoint the two equivalent scalar residuals are

$$\mathscr{R}(f, Z) = \mathsf{A} : (K + bH) - \tau_0 f - C, \tag{4.4}$$

$$\mathscr{N}(f, Z) = \frac{\mathscr{R}}{b} = \mathsf{A} : H - \frac{1}{b}(\tau_0 f + C - \mathsf{A} : K). \tag{4.5}$$

These are exactly `d:trace` and `e:normalized-trace`. The boundary values are $\tau_0 f = b_-$ on black inner faces, $\tau_0 f = b_+$ on white faces, and $f = 0$ on S_R. The scalar solve takes arbitrary bounded invariant Z with zero outer trace; it does *not* assume the floor $t \ge -\varepsilon$.

All homotopy stages. In the first stage the trace datum remains $F = h + C$, $h = \tau_0 f$, while $\lambda K(w, \cdot)$ is used in the second flux. Stage two adds a smooth collar term $D_0(x, w, h)$, nondecreasing in h. Stage three uses

$$F = h + (1 - \zeta)[C + D_0(x, w, h + \zeta b_{\mathrm{lift}})] + \zeta[\mathsf{A} : K + D_1(x, w)], \tag{4.6}$$

with inner height $(1 - \zeta)b_{\mathrm{lift}}$. Stage four retains the terminal trace solve $f = 0$ and scales the second source and boundary flux together to zero. In every stage $F_h \ge 1$ with the other arguments fixed. The auxiliary interpolation of `e:scalar-interpolation` is

$$[(1 - s)I + s\mathsf{A}] : H = (1 - s)\tau_0 f + s\, b^{-1}(F - \mathsf{A} : K), \qquad 0 \le s \le 1.$$

Its strictly positive height coefficient is not a claim that its gradient dependence is divergence form.

5 Complete Local Variations

All variations in this section fix g, K, n, ε, C and the domain. A metric, profile, moving-boundary, or parameter variation would need additional terms. Let $\delta f = \phi$, $\delta Z = \zeta$, $\delta P = \nabla\phi$, $\delta H = \operatorname{Hess}\phi$, and $p' = \mathrm{d}p/\mathrm{d}t$.

Proposition 5.1 (Smooth full first variation)**.** *On a bounded smooth coefficient range, the following are the exact first variations, including at $P = 0$:*

$$\delta t = \frac{4\zeta - b^2 P \cdot \delta P}{M}, \qquad \lambda_b := \frac{\delta b}{b} = pd\,\delta t - b^2 P \cdot \delta P, \tag{5.1}$$

$$\delta w = b\delta P + w\lambda_b, \qquad \delta v = 2w \cdot \delta w, \quad \delta d = -\delta v, \tag{5.2}$$

$$\delta p = p'\delta t, \qquad \delta M = v\delta p + p\delta v, \tag{5.3}$$

$$\delta\beta = \frac{M\delta p - (p+4)\delta M}{M^2}. \tag{5.4}$$

$$\delta \mathsf{A} = -\delta\beta\, w \otimes w - \beta(\delta w \otimes w + w \otimes \delta w), \tag{5.5}$$

$$\delta \log u = (p + 2 + pv)\delta t + b^2 P \cdot \delta P. \tag{5.6}$$

Consequently

$$D\mathscr{R}[\phi, \zeta] = b\mathsf{A} : \operatorname{Hess}\phi + \delta\mathsf{A} : (K + bH) + b\lambda_b \mathsf{A} : H - \tau_0\phi, \tag{5.7}$$

$$D\mathscr{N}[\phi, \zeta] = b^{-1} D\mathscr{R}[\phi, \zeta] - \mathscr{N}\lambda_b. \tag{5.8}$$

Dropping the last term in (5.8) is valid at an exact trace root, not at an arbitrary candidate. For the other stages replace $\tau_0\phi$ in (5.7) by the complete δF.

Proof. Differentiate D to obtain $\delta D/D = 2pv\delta t + 2b^2 P \cdot \delta P$. Differentiating $Z = t + \frac{1}{8}\log D$ gives the first formula. Since $\log b = \log l - \frac{1}{2}\log D$, its derivative is the second. The product, quotient, and tensor-product rules prove (5.2)–(5.6). Differentiating (4.4) and then dividing by the variable b proves (5.7)–(5.8). All denominators satisfy $D \geq 1$, $M \geq 4$, and $l > 0$, so these are smooth formulas at zero slope. For example there $\delta t = \zeta$, $\delta w = l\delta P$, $\delta\mathsf{A} = 0$, but $\delta b/b = p\zeta$ need not vanish. Smooth composition in the Hölder Banach algebras on a compact truncation proves that these pointwise variations are Fréchet derivatives in the strong spaces used below. □

For fixed Z, (5.8) has the form

$$\mathcal{L}\phi = \mathsf{A} : \operatorname{Hess}\phi + \mathfrak{b} \cdot \nabla\phi - c\phi, \qquad c = \tau_0/b > 0 \tag{5.9}$$

at the physical stage. The vector $\mathfrak{b}$ consists of *all* the δP terms in (5.1)–(5.8); it is not zero in general and depends on H, K, P, Z, p'. At other stages $c = \tau_0 F_h/b$ and the w- and A-dependence of F also enters $\mathfrak{b}$ and the Z derivative. Bounds on p' cannot be replaced by bounds on p alone.

6 Nondivergence, Weak Forms, and Green Equivalence

Proposition 6.1 (Drift-correct conversion)**.** *For a C^1 symmetric principal tensor and smooth ϕ,*

$$\mathcal{L}\phi = \operatorname{div}(\mathsf{A}\nabla\phi) + (\mathfrak{b} - \operatorname{div}\mathsf{A}) \cdot \nabla\phi - c\phi, \qquad (\operatorname{div}\mathsf{A})^j = \nabla_i \mathsf{A}^{ij}. \tag{6.1}$$

Writing $q = \operatorname{div}\mathsf{A} - \mathfrak{b}$, the weak Dirichlet form for $-\mathcal{L}$ on $V = H_0^1(\Omega_R)$ is

$$a(\phi, \psi) = \int_{\Omega_R} [\mathsf{A}\nabla\phi \cdot \nabla\psi + (q \cdot \nabla\phi)\psi + c\phi\psi], \mathrm{d}V_g. \tag{6.2}$$

If $\mathsf{A} \geq a_0 I$, $c \geq 0$, and $a_0 - C_P \|q\|_\infty =: \alpha_* > 0$, then a is coercive in the gradient norm and the weak inverse has norm at most α_*^{-1}. For prescribed trace with lift e_b, the actual inverse G gives $\phi = e_b + G(r - (-\mathcal{L})e_b)$. This is not a free-space kernel formula.

Proof. The product rule gives (6.1). Integrating the divergence term against a zero-trace test gives (6.2). Poincaré and Cauchy–Schwarz bound the drift term below by $-C_P \|q\|_\infty \|\nabla\phi\|_2^2$. The classical coercive-form argument gives the inverse and its norm. Subtracting the lift, applying the inverse, and reversing those operations proves both directions of the displayed equivalence. □

Neither symmetry nor coercivity of (6.2) is automatic from $\mathsf{A} > 0$. The small-drift condition is sufficient, not necessary: the classical maximum-principle/Schauder inverse used by the source is another route under smooth Dirichlet hypotheses. In particular, source Lemma `g:gradient` only assumes measurable χ; the conversion above is not justified there by differentiating χ.

An explicit negative control. On $0 < x_1 < 1$, take $\mathsf{A} = \operatorname{diag}((1+x_1)/2, 1, 1)$ and $f = x_1$. Then $\mathsf{A} : D^2 f = 0$, whereas $\operatorname{div}(\mathsf{A}\nabla f) = 1/2$. Its eigenvalues have exactly the axial pattern $(\chi, 1, 1)$ with $0 < \chi \le 1$. This is a test of operator conversion, not a physical exterior solution.

Axisymmetric coordinates are still three-dimensional. For a polarized metric $g = h + X^2 \mathrm{d}\varphi^2$ and invariant scalars, physical integration is $2\pi X \, \mathrm{d}V_h$, not bare meridional area. The axis is a coordinate degeneracy, not an added artificial boundary. For a flat cylindrical example $f = f(\rho, z)$, $\Delta f = f_{\rho\rho} + \rho^{-1} f_\rho + f_{zz}$. The angular Hessian contributes f_ρ/ρ even though $\partial_\varphi f = 0$. Smooth axis parity or a regular Cartesian chart must be retained in any panel method.

7 A Height Certificate Without a Floor

Theorem 7.1 (Unit residual bound in rescaled height)**.** *Fix the same continuous Z and smooth data on a compact truncation. Let f_* be a C^2 trace solution, and let f_c be any C^2 candidate, continuous to the boundary. Then at the physical endpoint*

$$\|\tau_0(f_c - f_*)\|_\infty \le \max\left\{ \|\tau_0(f_c - f_*)\|_{L^\infty(\partial\Omega_R)}, \|\mathscr{R}(f_c, Z)\|_\infty \right\}. \tag{7.1}$$

No floor assumption, gradient smallness, or coefficient-derivative bound is needed for this comparison. The same result holds for a homotopy trace $\mathsf{A} : (K + bH) = F$ whenever $F_h \ge 1$ and its other arguments agree when x, Z, P agree. This result is conditional on the existence of f_*.

Proof. If $f_c - f_*$ has a positive interior maximum, the two gradients agree. Hence their t, b, A agree. Their Hessian difference is negative semidefinite, while $b\mathsf{A}$ is positive definite. At that point

$$\mathscr{R}(f_c, Z) - \mathscr{R}(f_*, Z) \le -\tau_0(f_c - f_*).$$

The positive maximum is bounded by the residual divided by τ_0. At a negative interior minimum the reverse inequality gives the same absolute bound. Boundary extrema give the first term of (7.1). For the general datum integrate $F_h \ge 1$ between the two heights at the common x, Z, P. These arguments also apply to limiting suprema on the compact closure. □

The factor 1 is sharp over unconstrained boundary data: in the flat $K = C = 0$, $Z = 0$ example, constants $f_* = 0$, $f_c = a$ have residual $-\tau_0 a$. This sharpness is not a claim about a same-boundary nonconstant geometric example. In unscaled f the constant is $\tau_0^{-1} = e^{3n\varepsilon/2}$, so the estimate is not an n-uniform graph bound. A normalized residual $\mathscr{N}$ instead costs $\|b\mathscr{N}\|_\infty \le e \|\mathscr{N}\|_\infty$ in height. The unnormalized physical residual avoids this unnecessary multiplier.

Corollary 7.2 (Input sensitivity with a stated coefficient budget)**.** *Let f_i solve the physical trace problems for Z_i on the same domain. Suppose the segment $Z_s = Z_1 + s(Z_2 - Z_1)$ at f_2 stays in a bounded coefficient range, and define*

$$B_Z = \sup_{x,s} |\mathsf{A}_Z : (K + b \operatorname{Hess} f_2) + b_Z \mathsf{A} : \operatorname{Hess} f_2| \,. \tag{7.2}$$

Then

$$\|\tau_0(f_2 - f_1)\|_\infty \le \max\{\|\tau_0(f_2 - f_1)\|_{\partial\Omega_R,\infty}\,, B_Z \,\|Z_2 - Z_1\|_\infty\}. \tag{7.3}$$

Here $t_Z = 4/M$, $b_Z/b = 4pd/M$, and A_Z follows from (5.2)–(5.5) with $\delta P = 0$, $\zeta = 1$.

Proof. The fundamental theorem of calculus at fixed f_2 bounds $|\mathscr{R}(f_2, Z_1) - \mathscr{R}(f_2, Z_2)|$ by the stated quantity. The second residual vanishes. Apply Theorem 7.1 at Z_1. □

The displayed B_Z requires a Hessian bound and full profile derivatives. It is not a universal Lipschitz constant for the coupled exterior solve.

8 A Quantified Continuum Inverse Gate

For the source's axial principal matrix, an exact scalar optimization is

$$\gamma = \frac{2 + \chi}{2 + \chi^2}, \qquad \delta(\chi) = \|I - \gamma \mathsf{A}\|_{\mathrm{F}} = \frac{\sqrt{2}(1 - \chi)}{\sqrt{2 + \chi^2}}, \qquad \epsilon_C = \frac{\chi(4 - \chi)}{2 + \chi^2} > 0. \tag{8.1}$$

Indeed minimizing $3 - 2a(2 + \chi) + a^2(2 + \chi^2)$ over scalar a gives $a = \gamma$ and $\delta^2 = 1 - \epsilon_C$. Compared with unscaled defect $1 - \chi$, this gives a strictly smaller principal contraction defect for $0 < \chi < 1$. It need not improve the *unweighted* residual-to-solution constant, since γ also multiplies the residual. This is a classical Cordes specialization, not a new regularity theorem [6].

Theorem 8.1 (Full lower-order perturbation gate)**.** *Let $U \subset \mathbb{R}^3$ be bounded, convex, and $C^{1,1}$, and put $X = H^2(U) \cap H^1_0(U)$, $\|v\|_X = \|\Delta v\|_2$. Assume $\chi \ge \chi_0 > 0$ and let the Euclidean-coordinate realization of a frozen trace linearization be*

$$\mathcal{L}v = (\mathsf{A} + E) : D^2 v + \mathfrak{b} \cdot Dv - cv.$$

All coefficients are bounded; A has the axial eigenvalues above. Use constants C_1, C_0 such that $\|Dv\|_2 \le C_1 \|v\|_X$ and $\|v\|_2 \le C_0 \|v\|_X$. If

$$q_* := \|\delta(\chi)\|_\infty + \|\gamma E\|_{\infty,\mathrm{F}} + C_1 \|\gamma \mathfrak{b}\|_\infty + C_0 \|\gamma c\|_\infty < 1, \tag{8.2}$$

then $\mathcal{L} : X \to L^2(U)$ is a bounded isomorphism and

$$\left\|\mathcal{L}^{-1} r\right\|_X \le \frac{\|\gamma r\|_2}{1 - q_*}. \tag{8.3}$$

For invariant domain and coefficients, invariant forcing has an invariant solution. On a geometric chart E includes all metric principal errors, and $\mathfrak{b}$ includes the connection terms.

Proof. The Dirichlet Poisson inverse $P_\Delta : L^2 \to X$ exists by the classical convex-domain theory; $\|P_\Delta r\|_X = \|r\|_2$ and Miranda–Talenti gives $\|D^2 v\|_2 \le \|\Delta v\|_2$. Poincaré and integration by parts permit $C_1 = C_P$, $C_0 = C_P^2$. Solving $\mathcal{L}v = r$ is equivalent to

$$v = P_\Delta\{\gamma r + (I - \gamma(\mathsf{A} + E)) : D^2 v - \gamma \mathfrak{b} \cdot Dv + \gamma c v\}.$$

The linear part has norm at most q_*. Its geometric series is a two-sided bounded inverse of identity minus that part, proving existence, uniqueness and (8.3). Since γ is bounded above and away from zero, this solves precisely the original equation. Uniqueness and the group action give invariance. □

This gate concerns a whole infinite-dimensional Dirichlet space, not a matrix's smallest singular value. It does not cover arbitrary nonconvex exterior truncations by fiat. Local charts require localization errors and an assembly argument to yield a global inverse. The positive c helps the maximum principle but is treated by an absolute-value perturbation here; failure of (8.2) does not show that $\mathcal{L}$ is singular.

9 Ram-Equivalent Root and Residual Certificates

On a smooth compact Ω_R fix $0 < \alpha < 1$, the trace lift e_b, and a fixed invariant $Z \in C^{1,\alpha}$. Define

$$X_\alpha = \{v \in C^{2,\alpha}(\overline{\Omega}_R) : v|_{\partial\Omega_R} = 0,\ v \text{ invariant}\}, \qquad Y_\alpha = C^{0,\alpha}_{\text{inv}}(\overline{\Omega}_R),$$

and $\mathcal{F}(v) = \mathscr{R}(e_b + v, Z)$. The derivatives in Section 5 give $\mathcal{F} \in C^1$ on bounded open ranges, with a locally Lipschitz derivative when the profiles have the required next derivatives. The boundary realization is part of X_α.

Theorem 9.1 (A conditional local root, not merely a finite solve)**.** *Let $B : Y_\alpha \to X_\alpha$ be a bounded isomorphism. For a closed ball $\overline{B}(c, r)$ inside the coefficient domain suppose the* continuum *bounds*

$$\|B\mathcal{F}(c)\| \le \eta, \quad \|I - BD\mathcal{F}(c)\| \le \kappa_0, \quad \|B\| \le \mathfrak{B}, \quad \operatorname{Lip}(D\mathcal{F}) \le L_*, \tag{9.1}$$

hold and

$$q = \kappa_0 + \mathfrak{B}L_* r < 1, \qquad \eta + qr \le r.$$

Then the ball contains a unique trace root and its distance to c is at most $\eta/(1-q)$. No numerical candidate meeting these hypotheses for the M513 geometric data is supplied in this report.

Proof. $T(v) = v - B\mathcal{F}(v)$ has derivative norm at most q throughout the ball, by the line-segment formula and (9.1). Thus it contracts and maps the ball into itself. Completeness gives a unique fixed point. Injectivity of B makes it a root, and every root is a fixed point. Finally $\|v_* - c\| \le \eta + q\|v_* - c\|$ proves the estimate. □

The inverse of a smooth Dirichlet linearization, if quantitatively bounded in these Hölder spaces, can provide B. The H^2 inverse of Theorem 8.1 does not automatically give such a Hölder bound. Alternatively that theorem may be used in a compatible H^2 setting only after the nonlinear map and its derivative are proved bounded there. In dimension three, H^2 alone does not control ∇f in L^∞; no such nonlinear assumption is suppressed here.

For bounded coordinate isomorphisms $C_X : X_\alpha \to \mathcal{X}$ and $C_Y : Y_\alpha \to \mathcal{Y}$, define the exact Ram-represented equation

$$\widehat{\mathcal{F}}(a) = C_Y\mathcal{F}(C_X^{-1}a), \qquad D\widehat{\mathcal{F}}(a) = C_Y D\mathcal{F}(C_X^{-1}a)C_X^{-1}. \tag{9.2}$$

Injectivity proves root equivalence, and the chain rule proves the derivative. The preconditioner is $C_X B C_Y^{-1}$; transported norms preserve the same contraction constants. In untransported norms the condition numbers must be included. A redundant frame must be restricted to its consistent coefficient range. A finite panel bank is not an isomorphism on these unrestricted function spaces.

For a boundary-realized inverse with a legitimate kernel, localization is the substitution $y = x - a$ over $\{a : x - a \in \Omega_R\}$, with the appropriate physical density. It does not erase that support, the boundary lift, the diagonal singularity, the drift in (6.1), or unresolved coefficient tails. Finitely many zero moments cannot bound $\|\mathscr{R}\|_\infty$ or a full dual norm without a remainder theorem.

10 The Divergence Companion and Full Coupling

At the physical endpoint the companion is genuinely in divergence form:

$$V = u\mathsf{A}(4\nabla Z + K(w,\cdot)), \qquad \operatorname{div} V = u\Xi, \qquad \Xi = \delta_0\mathcal{T} + \rho + \delta_0\tau_0 b|P|^2 - m_0(t)C_n(\rho_0 + v\rho_1). \tag{10.1}$$

Notice $a\sigma = b|P|^2$ is smooth at zero slope; separately differentiating $|P|$ is unnecessary. Inner data use the source normal ν toward the exterior and

$$V_\nu = u\mathcal{B}, \qquad \mathcal{B} = -H_B + w_\nu(F - P_B) - N_B d, \quad P_B = \operatorname{tr}_B K,$$

while $Z = 0$ on the nonempty outer Dirichlet face. The outward normal of Ω_R on the inner boundary is $-\nu$.

Proposition 10.1 (Frozen second solve with the correct sign)**.** *Freeze a smooth input and its scalar solution, set $C_Z = 4u\mathsf{A}$, $d_0 = u\mathsf{A}K(w,\cdot)$, $s_0 = u\Xi$, $g_B = u\mathcal{B}$, and let $V_R = \{v \in H^1(\Omega_R) : v|_{S_R} = 0\}$. Then the frozen output z satisfies*

$$\int C_Z\nabla z\cdot\nabla\psi = -\int s_0\psi - \int d_0\cdot\nabla\psi - \int_B g_B\psi \qquad (\psi \in V_R). \tag{10.2}$$

If $u \ge u_0 > 0$, $\chi \ge \chi_0 > 0$ and the loads are bounded functionals, this has a unique weak solution with inverse norm at most $(4u_0\chi_0)^{-1}$ in the gradient norm. For a candidate z_c with the same outer trace, its functional residual r_c obeys

$$\|\nabla(z_c - z)\|_2 \le \frac{\|r_c\|_{V_R'}}{4u_0\chi_0}. \tag{10.3}$$

Proof. Integrate $\operatorname{div}(C_Z\nabla z + d_0) = s_0$ against ψ. The outer boundary term is zero and the inner outward flux is $-g_B$, which gives (10.2). The nonempty Dirichlet face supplies a Poincaré inequality; the principal form is coercive with the displayed constant. Lax–Milgram gives existence and uniqueness. Subtract the equations and test by the error to prove (10.3). There is no pure-Neumann compatibility condition. □

The source constructs a compact map by solving this *frozen* problem after the scalar solve. Freezing is an iteration, not the full derivative of the coupled equations. In particular

$$\begin{aligned}
\delta V &= \delta u\,\mathsf{A}(4\nabla Z + K(w,\cdot)) + u\delta\mathsf{A}(4\nabla Z + K(w,\cdot)) \\
&\quad + u\mathsf{A}(4\nabla\zeta + K(\delta w,\cdot)), && (10.4)\\
\delta(\operatorname{div} V - u\Xi) &= \operatorname{div}\delta V - \delta u\,\Xi - u\delta\Xi, && (10.5)\\
\delta\Xi &= \delta_0\delta\mathcal{T} + \delta_0\tau_0 b[2P\cdot\delta P + |P|^2\lambda_b] \\
&\quad - C_n[m_0'\delta t(\rho_0 + v\rho_1) + m_0\rho_1\delta v]. && (10.6)
\end{aligned}$$

The fixed-background boundary derivative is $\delta(V_\nu - u\mathcal{B}) = \delta V_\nu - \delta u\mathcal{B} - u\delta\mathcal{B}$, where $\delta\mathcal{B} = \delta w_\nu(F - P_B) + w_\nu\delta F - N_B\delta d$. One must use the same off-solution choice of F throughout an implementation. For the tensor definition of $\mathcal{T}$, the source specifies the *actual* trace $F_a = \mathsf{A} : (K + bH)$; replacing it by $\tau_0 f + C$ off a trace root changes the residual map. Both choices agree only when the trace equation is imposed. Appendix A gives its full variation.

The source's quadratic Hessian term is not automatically a bounded $H^1 \to H^{-1}$ nonlinearity. Its local gradient/Morrey estimates and bootstrap are genuine continuum obligations. Neither Proposition 10.1 nor a product of two finite inverses proves invertibility or contraction of the fully coupled nonlinear system.

11 Physical Branch and Quantified Perturbations

Write $x = A/(4\pi) > 0$, $q = Q_e^2 + Q_b^2 \geq 0$, $j = J$, and $s = \sqrt{q^2 + 4j^2}$. Then

$$P(x, q, j) = \frac{1}{4}\left(x + 2q + \frac{q^2 + 4j^2}{x}\right), \qquad x \geq s. \tag{11.1}$$

This notation q is unrelated to the weak-form drift.

Theorem 11.1 (Branch, equality algebra, and extremal stability)**.** *On $x \geq s$,*

$$P_x = \tfrac{1}{4}(1 - s^2/x^2) \geq 0, \qquad P_{xx} = s^2/(2x^3), \tag{11.2}$$

$$P(x, q, j) - \frac{q + s}{2} = \frac{(x - s)^2}{4x}. \tag{11.3}$$

For $s > 0$, $m^2 \geq P(x, q, j)$ on the physical branch is equivalent to

$$m^2 \geq \frac{q + s}{2}, \qquad s \leq x \leq 2m^2 - q + 2\sqrt{m^4 - m^2 q - j^2}. \tag{11.4}$$

If $0 \leq P - (q + s)/2 \leq \epsilon$, then

$$0 \leq x - s \leq 2\epsilon + 2\sqrt{\epsilon(s + \epsilon)}. \tag{11.5}$$

For $s = 0$, necessarily $q = j = 0$, and $P = x/4$; $x = 0$ is outside the positive-area class.

Proof. Differentiate (11.1) and complete the square to obtain (11.2)–(11.3). Multiplication of $P \leq m^2$ by $4x$ gives the quadratic $x^2 + (2q - 4m^2)x + s^2 \leq 0$. Its roots are $x_\pm = 2m^2 - q \pm 2\sqrt{m^4 - m^2 q - j^2}$. The minimum on the branch is $(q + s)/2$, attained at s, and $x_- x_+ = s^2$, so once that minimum is affordable the lower root is at most s and the upper root is at least s. This proves the equivalence. Writing $y = x - s \geq 0$ in (11.3) gives $y^2 \leq 4\epsilon(s + y)$; solving that quadratic proves (11.5). The zero-charge, zero-rotation statement is direct. □

If $x \geq s + \delta$ with $s > 0$, the inverse area sensitivity is bounded by $4/[1 - (s/(s + \delta))^2]$ via (11.2). This degenerates as $\delta \downarrow 0$; (11.5) correctly has square-root rather than uniform linear sensitivity there. For $j = 0$ the bound reduces to the charged expression $(x + q)^2/(4x)$; for $q = 0$ it becomes $x/4 + j^2/x$ with branch $x \geq 2|j|$.

Proposition 11.2 (Certified parameter boxes)**.** *Suppose a candidate (x_0, e_0, j_0), $e_0 = (Q_{e,0}, Q_{b,0})$, has $|x - x_0| \leq a$, $|e - e_0| \leq e_*$, $|j - j_0| \leq j_*$ and $x_- = x_0 - a > 0$. Set $q_0 = |e_0|^2$, $d_q = 2|e_0|e_* + e_*^2$, $q_+ = q_0 + d_q$, $j_+ = |j_0| + j_*$, and $s_0 = \sqrt{q_0^2 + 4j_0^2}$. The sufficient whole-box physical-branch condition is*

$$x_0 - a \geq s_0 + \sqrt{d_q^2 + 4j_*^2}. \tag{11.6}$$

Without assuming that condition, the following error bound still holds on the positive-area box:

$$\begin{aligned} |P(x, |e|^2, j) - P(x_0, q_0, j_0)| \leq{}& \tfrac{1}{4}\left(1 + \frac{q_+^2 + 4j_+^2}{x_-^2}\right) a \\ &+ \left(\tfrac{1}{2} + \frac{q_+}{2x_-}\right) d_q + \frac{2j_+}{x_-} j_*. \end{aligned} \tag{11.7}$$

If a certified lower mass bound is $m \geq m_- > 0$ and the right side is Δ_P, then $m_-^2 \geq P(x_0, q_0, j_0) + \Delta_P$ certifies the numerical inequality for those already validated data*. It does not validate their constraints or produce a Penrose theorem.*

Proof. $||e|^2-|e_0|^2| \le d_q$. Apply the reverse triangle inequality in $\mathbb{R}^2$ to $(q, 2j)$ and $(q_0, 2j_0)$, obtaining $|s - s_0| \le \sqrt{d_q^2 + 4j_*^2}$; this proves (11.6). Differentiate P with respect to x, q, j: $P_q = 1/2 + q/(2x)$, $P_j = 2j/x$, and use (11.2). Bound these derivatives on the straight segment in (x, q, j), which stays in the positive-area box. Integrating proves (11.7). The final assertion follows by ordering the two certified bounds. □

Model scale and a finite-check limit. The source sets $F_*(R) = \sqrt{P(R^2, q, j)}$, $a_* = -j/F_*$ and $b_* = (R^4 - s^2)/(4F_*R^2) > 0$, with $F_*'(R) = b_*/R$. Its sourced comparison contains

$$E_g \ge F_*(R) + \frac{b_*}{2} \log \frac{L_{\text{cross}}}{2R^2}.$$

For $L_{\text{cross}} = A/(2\pi)$ choose $R^2 = x$ on the strict branch. At $x = s > 0$ let $R^2 \downarrow s$; the logarithm tends to zero and $b_* \to 0$. This is only the algebraic final limit once the continuum comparison has been proved. A finite grid of R values is not that proof. Likewise the Kerr–Newman equality check $x = (m + \sqrt{m^2 - q - j^2/m^2})^2 + j^2/m^2$ is an algebraic consistency test, not original-data rigidity.

12 Continuum Obligations That Remain

The local results have no hidden global conclusion. The source's proof still needs the following specific links.

1. **Scalar solve without a floor.** In `e:trace-solve`, bounded input Z and large σ give $t \to -\infty$ and $e^{8Z} = e^{8t} + e^{(2n+8)t}\sigma^2$. Thus $d, \chi \asymp \sigma^{-8/(n+4)}$ and $b^{-1} \asymp \sigma$. The source uses these bounds for logarithmic boundary barriers and a two-point interior Lipschitz argument. Theorem 7.1 alone does not supply this gradient bound or existence.

2. **Quadratic source regularity.** The source's `g:gradient` proves gradient Hölder and Hessian Morrey control without differentiating measurable χ. Its cancellation identity $\mathsf{A}\nabla(|\nabla f|^2/2) - G_0\nabla f = (\mathrm{Hess}\, f - (\Delta f)g)\nabla f$ underlies the argument. The reflected face, drop-or-affine step, improvement of flatness, and the coupled bootstrap require continuum estimates. A frozen Cordes inverse proves none of those nonlinear steps by itself.

3. **Floor, ceiling, and compact map.** The source's `d:floor`, `e:ceiling`, `g:regularity`, and `e:existence` must supply uniform fixed-point bounds over all four homotopy stages and prevent boundary fixed points in the moving degree domain. The scalar inverse and mixed coercivity are only inputs.

4. **Exhaustion and ADM flux.** At fixed prepared data, fixed ε, and fixed n, first let $R \to \infty$ with estimates on every compact set including inner faces. The source then needs f exponentially decaying, $Z, t = O_2(r^{-1})$, integrable $\operatorname{div} V$, and

$$E_{\hat{g}} = E_g - \frac{\mathfrak{F}_n}{8\pi}, \qquad \mathfrak{F}_n = \int_{\Omega_{\text{red}}} u\Xi + \int_B V_\nu.$$

   Only afterward let $n \to \infty$ in this scalar energy comparison. The penalty estimate uses $\rho_1^2 \asymp r^{-4\gamma_1}$, $\gamma_1 > 3/4$, and the factors ℓ, $\tau_0/\ell = \ell^{1/2}$, $\ell^2/\tau_0 = \ell^{1/2}$ against polynomial losses. Unspecified fixed-n regularity constants cannot be substituted into that polynomial absorption.

5. **Sourced geometry and global comparison.** In `d:row-completion`, keeping the three potential rows fixed gives $|D_i|_g^2 + s_i^2 + 2s_iD_i(w) = |D_i|^2_{g+l^2\mathrm{d}f^2} + (s_i + D_i(w))^2$. Axis regularity,

strict preparation, the conformal cut, crossing-length control and Riemannian comparison remain necessary. Then $\varepsilon \downarrow 0$ and preparation removal follow in the stated order.

6. **Equality in the original data.** The source's adjoint, boundary lapse dichotomy, quotient identification, spacetime lift, and smooth horizon attachment in `adjoint.tex` and `rigidity.tex` are separate obligations. Vanishing residual or equality in (11.3) is not a stationary adjoint or a spacetime embedding.

No interchange of these limits, no convergence of deformation metrics as $n \to \infty$, and no extension beyond the stated topology is inferred here.

13 Benefits, Costs, and Future Research

Benefits and advantages. The complete variations in Section 5 and Appendix A make a trace or coupled residual implementation auditable: the implicit t, profile derivatives, off-root normalization, and actual-trace dependence can each be checked separately. The unit rescaled-height bound in Theorem 7.1 provides a useful conditional error measure without imposing the floor on the scalar solve. The drift-correct conversion prevents a divergence-form discretization from silently changing the trace equation. Conserved total angular flux and the parameter boxes in Proposition 11.2 provide branch safety: electromagnetic angular momentum and a certified area margin cannot be replaced by their bare or sampled counterparts.

Costs and disadvantages. These benefits require bounds on the true residual and coefficient range, not only finite moments. The unscaled graph constant $\tau_0^{-1} = e^{3n\varepsilon/2}$ grows rapidly, B_Z requires Hessian and profile-derivative control, and the inverse gate deteriorates as $\chi_0 \downarrow 0$ or $q_* \uparrow 1$. Convex-chart estimates do not automatically assemble on the nonconvex exterior; boundary lifts, axis parity, metric weights, and the prescribed decay all have costs. Near extremality, the area error is only square-root controlled. Validated quadrature, unresolved modes and end tails would add work to any numerical method. No speedup, lower total cost, or better nonlinear convergence rate has been demonstrated against an existing solver.

Further improvements and credible next steps. At fixed (n, ε, R), a concrete next experiment is an invariant conforming candidate with the assigned inner and outer data. Interval enclosures of the physical residual, t, χ, and the complete Jacobian could then be combined with a proved complement-space inverse and a compatible nonlinear norm to test Theorem 9.1. For the coupled problem this requires all terms in (10.5), including boundary and quadratic Hessian terms, rather than just the two frozen solves. A useful analytical improvement would replace the absolute lower-order costs in (8.2) by an estimate exploiting the positive height term and actual drift structure. Its success must be measured by a proved full-operator bound, not by the principal Cordes defect alone. These are proposed steps, not achieved certifications.

Extensions and their obstacles. Passing a bounded-domain certificate toward infinity requires radius-uniform estimates and explicit ADM and electromagnetic tail errors, in the source's stated order of limits. Allowing charged matter would require the flux defects of Proposition 3.1 and a new compatible sourced comparison; conservation cannot simply be retained. Multiple boundary components or different topology require additional period and gauge data, revised axis-potential normalizations, and new cut/comparison arguments. Removing the assumed physical-area condition requires an applicable geometric area theorem, not branch algebra. Extremal equality needs separate

horizon or cylindrical-end analysis; the sensitivity estimate (11.5) supplies no rigidity. These future research directions do not alter the demonstrated C2/5 contribution or close the global obligations in Section 12.

14 Private Passage Locators and Evidence Discipline

The actual routing files are `_audit/corpus-core-map.txt`, `_audit/corpus-foundations-map.txt`, and `_audit/corpus-applications-map.txt`. Core E3/E4/E5 delimit boundary closure, nonlinear variation, and physical weak forms; foundations F9/F10 route to the three proof sources below; applications C8 confirms the residual-certificate use case. Routing entries are not proof evidence. The cited proofs themselves were read locally:

Astra TR8 [7]. `sec:elliptic`, `thm:coercive`, `eq:operator-domain`, `eq:compatibility`, and `thm:green-equivalence`, source lines 81–160; the semilinear equivalence and invariant-ball limitation, lines 270–293. The inverse is the actual boundary-realized operator, and coercivity is a hypothesis, not a consequence of a transform.

Astra TR6 [8]. Theorem "Full Urysohn variation", `eq:du`, `eq:ddu`, and `eq:newtonfull`, lines 329–384. Both source-state and observation-state terms are differentiated. The finite-dimensional fundamental theorem of calculus under uniform derivative bounds is the proof mechanism transferred to our coefficient compositions.

Astra TR1 stability [9]. `thm:ball`, `eq:radii`, lines 138–171, and `prop:transfer`, lines 195–208. These prove the actual Banach self-map/root argument and bounded-coordinate equivalence, while excluding a merely rectangular left inverse as a root certificate. Exact full-file SHA-256 values, line ranges, passage hashes and source assembly records are supplied in `_audit/M513-provenance.json` and the freeze manifest. Hashes identify bytes; they do not establish truth. No source originals were modified or uploaded.

15 Contribution Assessment and Validation Scope

The rating is **C2/5**: a little contribution, not significant under the study's contribution scale. The source-specific realized results are the smooth coefficient and residual differentiation (Proposition 5.1), the unit rescaled-height residual and input-sensitivity bounds (Theorem 7.1 and Corollary 7.2), the explicitly limited full-operator inverse gate (Theorem 8.1), and parameter-box and extremal sensitivity certificates (Theorem 11.1, Proposition 11.2). The charge/total-angular-flux defect formula and frozen mixed residual estimate retain the source's conservation and boundary signs. These are transparent classical specializations and implementation safeguards. They do not establish a notable global advance, a repair of a demonstrated source gap, a stronger Penrose inequality, or external novelty.

Reproducible checks

The companion exact check script tests rational identities for the implicit variation, smooth zero-slope limits, trace normalization, the axial eigenvalues, the square completion and its derivative, the nondivergence negative control, branch polynomials, model parameters, flux signs, and parameter budgets. It also checks source assembly, author lines, abstract rating, and resolved LaTeX labels.

Results and precise scopes are in `_audit/M513-checks.json`. Symbolic identities verify algebra; finite rational examples only test those examples. Neither kind supplies geometric existence, an interval-certified continuum residual, a full-space nonlinear inverse for the coupled problem, nor a proof of the source's global theorem.

The complete scientific TeX was saved before testing, PDF export, or provenance production for parent proof review. Native editor/compiler dispatch status, physical export status, all-page visual inspection and the final path/SHA-256 verification are recorded in the M513-owned audit files. The export uses the existing helper, with at most two formatting rounds. Delivery status and contribution rating remain separate facts.

A Variation of the Smooth Tensor Source

For reproducible full coupling write $T = \nabla t$, $S_1 = K + bH$, $F_a = \mathsf{A} : S_1$, and

$$Q_1 = S_1 + p(\mathrm{d}t \otimes w^\flat + w^\flat \otimes \mathrm{d}t), \quad \mathfrak{B}(A, B) = A : B - (\operatorname{tr} A)(\operatorname{tr} B), \quad P_A = A - (\operatorname{tr} A)g.$$

Set $A_d = I - w \otimes w$ and $W = \beta S_1(w, w)$. The source's `d:quadratic-tensor` is the smooth identity

$$\mathcal{T} = \tfrac{1}{2}\mathfrak{B}(Q_1, Q_1) + 2P_{Q_1}(w, T) + 4(p+1)A_d(T, T) + F_a[F_a + W + 2(p+1)w \cdot T]. \tag{A.1}$$

It avoids a choice of axis at $P = 0$. Define

$$\begin{aligned}
\delta S_1 &= b \operatorname{Hess} \phi + b\lambda_b H, \qquad \delta F_a = \delta\mathsf{A} : S_1 + \mathsf{A} : \delta S_1, \\
\delta Q_1 &= \delta S_1 + \delta p(\mathrm{d}t \otimes w^\flat + w^\flat \otimes \mathrm{d}t) \\
&\qquad + p(\mathrm{d}\delta t \otimes w^\flat + w^\flat \otimes \mathrm{d}\delta t + \mathrm{d}t \otimes \delta w^\flat + \delta w^\flat \otimes \mathrm{d}t), \\
\delta W &= \delta\beta S_1(w, w) + \beta\delta S_1(w, w) + 2\beta S_1(\delta w, w).
\end{aligned}$$

Then the full derivative, at fixed background, is

$$\begin{aligned}
\delta\mathcal{T} = {} & \mathfrak{B}(Q_1, \delta Q_1) + 2P_{\delta Q_1}(w, T) \\
& + 2P_{Q_1}(\delta w, T) + 2P_{Q_1}(w, \nabla\delta t) \\
& + 4\delta p\, A_d(T, T) + 8(p+1)[A_d(T, \nabla\delta t) - (w \cdot T)(\delta w \cdot T)] \\
& + \delta F_a[2F_a + W + 2(p+1)w \cdot T] \\
& + F_a[\delta W + 2\delta p\, w \cdot T + 2(p+1)(\delta w \cdot T + w \cdot \nabla\delta t)].
\end{aligned} \tag{A.2}$$

Proof. Differentiate each of the four terms in (A.1). Symmetry of $\mathfrak{B}$ yields its first term. The three entries of $P_{Q_1}(w, T)$ each vary. Since $\delta A_d = -\delta w \otimes w - w \otimes \delta w$, the third summand gives the terms involving δp and A_d in (A.2). The product rule for the final summand gives the last two lines. These formulas, (10.6), and the boundary derivative retain every first-order term; $\nabla\delta t$ includes derivatives of the implicit coefficients in (5.1). □

References

[1] OpenAI. *The Kerr–Newman Penrose Inequality for Axisymmetric Electrovacuum Exteriors.* October 5, 2026. Pinned source assembly, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public pinned directory.

[2] M. Khuri, B. Sokolowsky, G. Weinstein. *A Penrose-Type Inequality with Angular Momentum and Charge for Axisymmetric Initial Data*. General Relativity and Gravitation 51, 118 (2019). arXiv:1902.00501v2, Theorem 1.1 and Corollary 1.2.

[3] M. E. Gabach Clement, J. L. Jaramillo, M. Reiris. *Proof of the area–angular momentum–charge inequality for axisymmetric black holes*. Classical and Quantum Gravity 30, 065017 (2013). arXiv:1207.6761.

[4] X. Feng, R. Yan, S. Gao, Y.-K. Lau, S.-T. Yau. *Geometric Inequality for Axisymmetric Black Holes With Angular Momentum*. Classical and Quantum Gravity 42, 065022 (2025). arXiv:2312.10590v2.

[5] D. Xu. *The Angular Momentum Penrose Inequality*. arXiv:2512.06918v4, January 2, 2026. Current preprint claim, not independently certified here.

[6] I. Smears, E. Süli. *Discontinuous Galerkin finite element approximation of nondivergence form elliptic equations with Cordès coefficients*. SIAM Journal on Numerical Analysis 51 (2013), 2088–2106. Author-hosted article, Section 2, Lemma 1 and Theorems 2–3.

[7] M. Subbarao. Confidential Astra IE/PDE series, TR8, `TR8-lin-non-lin.tex`. Local path: `C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Exact passages specified in Section 14.

[8] M. Subbarao. Confidential Astra IE/PDE series, TR6, `TR6-lin-non-lin.tex`. Local path: `C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR6-lin-non-lin/TR6-lin-non-lin.tex`. Full Urysohn variation and full Newton equation.

[9] M. Subbarao. Confidential Astra IE/PDE series, TR1, `stability.tex`. Local path: `C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`. Preconditioned nonlinear certificate and transfer of nonlinear equations.